

Illustration by SIDDHANT JUMDE

tories, enhances agricultural household purchasing power, stimulates the rural economy, and aids climate change mitigation. Horticulture, which encompasses fruits, vegetables, flowers, plantations, spices and medicinal plants, has been a key diversification domain since the 1980s. In 2023-24, horticulture output was 359 MT from around 28 million hectares, exceeding the foodgrains output of 329 MT. Research shows that small and marginal farmers benefit more from diversification into fruits and vegetables, particularly the latter. A National Sample Survey Office (NSSO) survey revealed that additional income due to an area shift to horticulture was as high as Rs 69,129 per ha. in 2012-13. This shows that diversification towards horticulture can result in higher incomes for farmers.

Potential in horticulture: Horticultural crops genetically have more potential than the agronomic, and can yield over 30 tonnes per hectare. Current productivity averages are low, indicating growth potential through genetic improvement and better cultivation practices. Among these:

- **Water usage:** Fruits, vegetables, flowers and some spices are high water-demand crops. Practices like micro-irrigation boost productivity (25-50 per cent) and save water
- **Food loss and waste:** Minimising food losses (4.5-15.9 per cent) through strengthened agriculture logistics and processing is crucial. This will also help in transferring greater income to the farmers
- **Market access and trade:** Efficient marketing and transparent price discovery mechanisms are needed. Enhancing agri-logistics and cold storage infrastructure is vital
- **Agri-processing:** Right now it is low (12 per cent in the organised sector), particularly in fruits and vegetables (2.5 per cent). Upgrading capabilities is essential for market integration

Other challenges/ thrust areas: The capital to output ratio in horticulture is relatively high, but the output per unit of land and its monetised value are higher too. Supply-side constraints must be addressed to respond to demand growth. Some focus areas are:

- Address structural challenges via FPOs (farmer producer organisations), etc. to improve operational efficiency of small farms
- Risk management through contract farming, real-time advisories and crop insurance
- Maximise production through orchard rejuvenation, rational crop geometry, besides promoting alternative technologies like hydroponics and aeroponics
- Digital technologies for precision farming, certified seed production and nursery banks
- Alternative marketing channels, export facilitation and intervention schemes akin to MSP
- Job generation and supplementary incomes through processing and other activities like beekeeping and mushroom farming

THE GIANT STEPS

- **In FY24, horticulture output was 359 MT from around 28 mha, outstripping foodgrains at 329 MT**
- **India's per capita per day consumption of fruits and vegetables is far below the ICMR benchmark of 92 gms and 300 gms respectively**
- **With our projected population of 1.65 billion by 2050, the country must target a horticulture output of 1,000 MT by 2047/50**

Green Revolution-plus: India's per capita per day consumption of fruits and vegetables is far below the ICMR and National Institute of Nutrition recommendation of 92 gms and 300 gms respectively. With our projected population of 1.65 billion by 2050, the country has to target a horticulture output of 600 MT by 2030, and 1,000 MT by 2047/50. The mission has to be anchored around science, technology and innovation and integrated horticulture value chain systems. Since the value chain takes off from the results of R&D, it is key to driving the horticultural revolution. The demands on R&D include productivity improvement through germplasm enhancement and its utilisation by focussing on documentation, characterisation and conservation of plant genetic material; identification and use of genes in hybridisation for resistance against climate change, pests and disease; application of biotechnology to tap diverse traits including resistance against biotic and abiotic stresses; enriching bioavailability of nutrients; high quantum generation of biomass that can strengthen bio-resource-based enterprises. Achieving 'Green Revolution-plus' through horticulture alongside productivity enhancements in staples is key to India's agricultural future. ■

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